

# Lee Seou

Industrial Engineering

## CONTACT

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github.com/Seou-Web

## SKILLS

### Tools

Python   MATLAB

### Numerical Methods

LU Factorization  
Gauss-Seidel   Runge-Kutta  
FDM   FFT  
Interpolation

### Data Analysis

ANOVA   Statistical Analysis

# Capstone Design Project

## Visualization Simulator for Cosmetic Manufacturing Vessels Using Manufacturing Data

### Smart Factory Capstone Design 1 / Team 12

Developed a Population Balance Model (PBM)-based simulator with multiple quality indicators to predict and judge scale-up emulsification quality in cosmetic emulsion manufacturing vessels (CEVSim).

- Responsible for results analysis and project management
- Analyzed droplet size distribution data to evaluate emulsion quality and predict manufacturing outcomes
- Project outcomes resulted in a pending patent application and a registered software work (see Research Outcomes below)

# Research Outcomes

## Patent Application Under Examination

Method and System for Predicting and Determining Scale-up Emulsification Quality of Cosmetic Emulsion Manufacturing Vessels Using a Population Balance Model and Multiple Quality Indicators

- Applicant: Sungkyunkwan University Industry-Academic Cooperation Foundation
- Inventors: Jeong Jongpil, Kang Mingyu, **Lee Seou**, Lee Hakseung
- Reference No.: R-2026-0578-KR-1

## Software Registration — CEVSim Registered

Cosmetic Emulsification Vessel Scale-Up Simulator — Computer Program Work, Application Software / Industrial S/W

- Authors: Jeong Jongpil, Yoo Jaekyung, Kang Mingyu, **Lee Seou**, Lee Hakseung
- Registration No.: C-2026-030853 · Created May 29, 2026

# Research Papers

## CFD simulation of a Rushton turbine stirred-tank using open-source software with critical evaluation of MRF-based rotation modeling

### Individual / Mar. 2026

Reid, A., Rossi, R., Cottini, C., Benassi, A. — *Meccanica* (Springer), 2025, pp. 1613–1637.

**BioSTEAM: A Fast and Flexible Platform for the Design, Simulation, and Techno-Economic Analysis of Biorefineries under Uncertainty**

Individual / May 2026

Cortés-Peña, Y. et al. — *ACS Sustainable Chemistry & Engineering*, 2020, 8(8), 3302–3310.

**A population balance model for cosmetic emulsion design: A multiscale approach**

Team 12 / May 2026

Calvo, F., Gómez, J. M., Ricardez-Sandoval, L., Álvarez, O. — *Chemical Engineering Science*, 2024, 287: 119737.

**Education**

**B.S. in Industrial Engineering — Sungkyunkwan University**

2025.03 ~ Present

Lee Seou — Capstone CV

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